Minn Bo Bo

Software Engineer | Full-Stack, Cloud & DevOps

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PROFESSIONAL SUMMARY

Software Engineer and primary developer of the three-platform MANGOs educational metaverse ecosystem, owning full-stack architecture, real-time communication, data systems, storage, and cloud infrastructure. Works across Node.js, TypeScript, React, PostgreSQL, REST APIs, AWS, Docker, and CI/CD, with a focus on maintainable delivery, security, and reliable operations. First author of two IEEE publications on authentication and tokenomics for interconnected metaverses.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, InterSystems ObjectScript, SQL

Backend: Node.js, Express.js, Flask, REST APIs, OAuth/SSO, WebSockets, Sequelize ORM, Azure Functions

Frontend: React, React Native, Redux Toolkit, HTML, CSS, SCSS

Databases: PostgreSQL, MongoDB, Microsoft SQL Server, InterSystems Caché

Cloud & DevOps: AWS, EC2, IAM, S3, Route 53, CloudWatch, Docker, Docker Compose, Nginx, Linux, GitHub Actions, CI/CD

Testing & AI: Jest, unit testing, integration testing, TensorFlow, TensorFlow.js, document processing

PROFESSIONAL EXPERIENCE

Software Engineer - Full Stack & DevOps
Aug 2024 - Present

Terrafirma | Software delivery for William Green, Australia

- Develop and maintain full-stack systems for William Green, an Australian dental equipment distribution and servicing company.
- Build TypeScript and Node.js backend services, REST APIs, and PostgreSQL data workflows for production applications.
- Support delivery and operations with AWS, Docker Compose, Nginx, Linux, GitHub Actions, and CI/CD practices.
- Use Jest, unit testing, and integration testing to protect backend behavior and improve release confidence.

Full-Stack Developer & DevOps Engineer
Oct 2023 - Aug 2024

iBeky | Medical services technology

- Led Node.js backend development for internal web and mobile projects using Express.js and Jest test suites.
- Provisioned AWS development environments and configured Nginx reverse proxies for scalable application delivery.
- Administered databases and disaster-response procedures to support data integrity, recovery, and availability.
- Researched AI modules and built Python utilities and Flask applications for document and data workflows.

Junior Full-Stack Developer
Jan 2023 - Oct 2023

Jonda Health | Singapore healthcare technology

- Served as principal backend developer for JondaX and guided interns through implementation and delivery.
- Developed Python-based AI utilities for document processing, data extraction, and text formatting.
- Integrated AWS services and contributed to frontend pages using JavaScript, HTML, CSS, and React Native.

Backend Developer & Database Designer
2021 - 2022

Australian learning management system startup

Designed database systems, secure routes, and permission models; developed REST-based Azure Functions with Node.js, Express.js, Sequelize ORM, and JavaScript.

EDUCATION

Master of Engineering, Electrical Engineering | GPA: 4.00/4.00 **2024 - 2026**
Chulalongkorn University, Thailand | Telecommunication Networking and Systems

BEng (Hons), Electrical and Electronic Engineering - First Class **2022 - 2024**
Liverpool John Moores University, United Kingdom

Electronic and Communication Engineering **2016 - 2022**
Yangon Technological University, Myanmar | Transferred

SELECTED ENGINEERING PROJECT

Primary Developer - MANGOs Educational Metaverse Ecosystem
MANGOs | School | Scene Studio | Chulalongkorn University
Mentored by Professor Lunchakorn Wuttisittikulkij

- Architected, built, and continue to own all application screens, frontends, backend services, real-time socket communication, APIs, data structures, databases, storage, deployment, and infrastructure across three integrated platforms.
- Built MANGOs identity and ecosystem services plus MANGOs School: cross-platform authentication, metaverse registry, social and marketplace services, organizations, course administration, progress tracking, video checkpoints, quizzes, certificates, bilingual UX, and public API integrations.
- Built Scene Studio's browser-based 3D workflows, including scene manifests, asset ingestion, draft and version publishing, interactions, decoration rules, Worlds, canvases, plots, and shared sessions; collaborated with specialists who create course content, 3D scenes and assets, and Unity metaverses.

SELECTED PUBLICATIONS

1. **M. B. Bo** et al., "Authentication and Identification System for Interconnected Metaverses: Consideration and Implementation," IEEE ECTI-CON, 2025. doi:10.1109/ECTI-CON64996.2025.11100953

   Proposed and implemented an authentication and identification system for interconnected metaverses, addressing identity management and interoperability across participating virtual environments.

2. **M. B. Bo** et al., "Design and Implementation of Tokenomics and Ecosystem Model for Interconnected Metaverses," IEEE CCISP, 2025, pp. 225-229. doi:10.1109/CCISP67522.2025.11282105

   Designed and implemented a tokenomics and ecosystem model for interconnected metaverses, defining a structured framework for participation, incentives, and value exchange.

SELECTED AWARDS

- **Third Prize, Huawei Developer Competition Asia Pacific** (2024)

  Developed a blockchain-powered academic publishing platform using Huawei Cloud services. The platform enables students and subject-matter experts to publish research and project work while supporting academic integrity, reputation building, and revenue opportunities. Competition result.

- **Merit Award, Asia Pacific ICT Alliance Awards** (2023)

  Led development of a two-stage deep learning model for lung segmentation and pulmonary tuberculosis detection from chest X-rays. The solution was integrated into a doctor-patient cloud ecosystem and received a merit award in the tertiary-level AI category. Official winner list.

- **First Prize, APT IoT Smart Challenge for Agri-Tech** (2022)

  Developed an AWS-backed cloud infrastructure and deep learning component for an Arduino-based, microcontroller-integrated smart greenhouse system in the senior-level agriculture technology competition.

CERTIFICATIONS & LANGUAGES

Certifications: DevOps on AWS Specialization; Deep Learning Specialization; DeepLearning.AI TensorFlow Developer Professional Certificate; TensorFlow: Advanced Techniques Specialization.

Languages: Burmese (native); English (IELTS 8, official test; Diploma in English, Wall Street English).